Bonu Tarun

B.Tech in Computer Science and Engineering
Gokaraju Rangaraju College of Engineering
Hyderabad, Telangana, India
2024 – 2028

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Portfolio

Professional Summary

CS undergraduate (CGPA 8.95/10) building ML pipelines (PCA, K-Means, BIRCH) and full-stack apps (React, Node.js, Socket.IO). 180+ LeetCode problems solved; 3rd place in a college hackathon. Targeting backend and ML engineering internships.

Education

Gokaraju Rangaraju College of Engineering *2024 – 2028*
B.Tech in Computer Science and Engineering CGPA: 8.95/10

Prathibha Junior College *2022 – 2024*
Telangana State Board of Intermediate Education Percentage: 98.1%

Personal Projects

Customer AI – Real-Time Customer Segmentation & Business Intelligence System *2026*
Python, Streamlit, Pandas, NumPy, Scikit-learn, PCA, K-Means, BIRCH GitHub | Live Demo

- Built an end-to-end ML pipeline (log transform, StandardScaler, PCA, K-Means) on the UCI Wholesale Customers dataset, retaining 92.07% of variance with three principal components.
- Benchmarked K-Means against BIRCH using silhouette, Davies-Bouldin, and Calinski-Harabasz metrics; K-Means scored 0.288 silhouette vs. 0.194 for BIRCH.
- Shipped a 5-page Streamlit dashboard with real-time segmentation, what-if simulation, business CSV classification, and PDF reports, backed by a 20-test Pytest suite.

SyncSound – Real-Time Synchronized Music Listening & Video Calling Platform *2026*
React 19, TypeScript, Node.js, Express, Socket.IO, WebRTC, Agora GitHub

- Architected a server-authoritative playback model in Node.js/Socket.IO where one room timeline drives synchronized play, pause, seek, and track changes, with revision handling and drift correction for late joiners and reconnecting clients.
- Integrated Agora WebRTC for private two-party video calls alongside shared YouTube playback, live chat with history replay, a shared queue, and rate-limited emoji reactions.

Varnam Bangles – Luxury Jewellery Storefront with GitHub-Backed Admin *2026*
Node.js, Express, EJS, GitHub Contents API, bcrypt, Helmet GitHub

- Built a full-stack storefront (Express/EJS) with a session-authenticated admin panel for product, category, and image CRUD, persisting JSON and images directly to a GitHub repository via the Contents API with SHA-based conflict retry.
- Hardened with CSRF protection, rate limiting, Helmet CSP, bcrypt auth, and server-side MIME/size validation.

Technical Skills

Languages: Java, Python, C, JavaScript, SQL
Core CS: DSA, OOP, DBMS, OS, Computer Networks
Frameworks & Data: React, Node.js, Express, Streamlit, Pandas, NumPy, Scikit-learn, PCA, K-Means, BIRCH
Tools & Databases: PostgreSQL, Git, GitHub, Postman, Swagger, Plotly, Socket.IO

Achievements

- Secured 3rd place in a college-level hackathon building an AI-powered anomaly detection and explanation system with RAG and locally hosted LLMs.
- Solved 180+ LeetCode problems; led a 3-member team to complete a second-year academic project.

Certifications

AWS Academy — Generative AI Foundations, ML Foundations, ML for NLP *2026*
EduSkills Academy — Generative AI, Deep Learning & Language Models *2026*
Oracle Academy — Database Programming with SQL *2026*